Online Appendix for

Causal Inference with Ordinal Outcomes: Flexible Estimation Framework Using Normalizing Flows

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This appendix provides supplementary theoretical derivations, estimation details, extensions to heterogeneous reporting thresholds, comprehensive simulation designs with full results, and detailed tables for the empirical applications.

Contents

A. Additional Theoretical Details

A.1. Probability-Based Estimands and Latent Scale Invariance

Recall that the observed ordinal response $Y_i \in \{0, 1, \ldots, J\}$ is generated via the threshold-crossing model:

$$Y_i = j \quad \Longleftrightarrow \quad \alpha_{j-1} < Y_i^* \leq \alpha_j, \tag{A.1}$$

where Y_i^* is the latent potential outcome and $-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$.

Let $\mathcal{T} : \mathbb{R} \to \mathbb{R}$ be a strictly increasing, positive affine transformation: $\mathcal{T}(y) = a + c\,y$ with $c > 0$. Under this transformation, we define the transformed latent outcome $\tilde{Y}_i^* = \mathcal{T}(Y_i^*)$ and the transformed thresholds $\tilde{\alpha}_j = \mathcal{T}(\alpha_j)$. The observed category probabilities under the transformed scale are given by:

$$\begin{aligned}
P(\tilde{Y}_i = j) &= P(\tilde{\alpha}_{j-1} < \tilde{Y}_i^* \leq \tilde{\alpha}_j) \\
&= P(a + c\alpha_{j-1} < a + cY_i^* \leq a + c\alpha_j) \\
&= P(\alpha_{j-1} < Y_i^* \leq \alpha_j) = P(Y_i = j).
\end{aligned}$$

Because the category probabilities $\pi_j(d) = P(Y_i(d) = j)$ remain unchanged under any positive affine scale shift, any causal quantity defined directly as a functional of these probabilities is invariant to the choice of scale.

A.2. Wasserstein Distance as a Distributional Metric

To summarize overall treatment effects across the entire distribution without relying on arbitrary numerical score assignments, we employ the one-dimensional L_1 Wasserstein distance (also known as the Earth Mover's Distance). For two cumulative distribution functions (CDFs) F_1 and F_0 defined over the ordered categories $\{0, 1, \ldots, J\}$, the Wasserstein distance is formulated as:

$$\mathcal{W}(F_1, F_0) = \sum_{j=0}^{J-1} \left| F_1(j) - F_0(j) \right|, \tag{A.2}$$

where $F_d(j) = P(Y_i(d) \leq j)$ for $d \in \{0, 1\}$. This metric measures the minimum cost of shifting probability mass to transform the control outcome distribution into the treatment outcome distribution, preserving the natural ordering of categories.

B. Extensions: Heterogeneous Reporting Thresholds

The threshold-crossing model in Section **??** assumes that all respondents share common thresholds α_j. In many survey settings, respondents may interpret the ordinal response scale differently based on their individual characteristics, a phenomenon known as differential item functioning (DIF).

We can extend the proposed flexible estimation framework to accommodate heterogeneous reporting behavior. Following the generalized ordered response framework (**?**), we model the individual-specific thresholds $\alpha_{j,i}$ as a function of respondent-level covariates $W_i \subseteq X_i$:

$$\alpha_{j,i} = \alpha_j + W_i^\top \gamma_j, \tag{B.1}$$

where γ_j is a vector of parameters capturing how the thresholds shift across individuals, subject to the ordering constraint:

$$\alpha_{0,i} < \alpha_{1,i} < \cdots < \alpha_{J-1,i} \quad \forall\, i. \tag{B.2}$$

Under this specification, the probability of observing category j becomes:

$$\begin{aligned} P(Y_i = j \mid D_i, X_i) = F_\phi &\left(\alpha_j + W_i^\top \gamma_j - h(X_i) - D_i^\top \tau \mid D_i, X_i\right) \\ -F_\phi &\left(\alpha_{j-1} + W_i^\top \gamma_{j-1} - h(X_i) - D_i^\top \tau \mid D_i, X_i\right). \end{aligned} \tag{B.3}$$

This generalized formulation is fully compatible with both the semiparametric KDE-based estimator and the conditional normalizing flow framework. In the flow-based setting, the parameters γ_j are jointly estimated via maximum likelihood alongside the flow parameters ϕ and the index coefficients.

C. Estimation Details and Proofs

C.1. KDE-Based Ordinal Regression

This appendix provides details for the semiparametric KDE-based estimator. We first define the quasi-likelihood, then describe the kernel estimation of the error distribution and state a consistency result.

C.1.1. Model and Quasi-Likelihood

Recall the latent outcome model:

$$Y_i^* = f(X_i, \beta) + D_i^\top \tau + \varepsilon_i,$$

and the threshold-based reporting function:

$$Y_i = j \quad \Longleftrightarrow \quad \alpha_{j-1} < Y_i^* \leq \alpha_j, \qquad j = 0, 1, \ldots, J,$$

with $-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$. For notational simplicity, define the latent outcome index:

$$V_i(\beta, \tau) = f(X_i, \beta) + D_i^\top \tau.$$

Let F denote the (unknown) CDF of ε_i. Conditional on (X_i, D_i), the probability of observing category j is:

$$p_j(X_i, D_i; \alpha, \beta, \tau, F) = F\big(\alpha_j - V_i(\beta, \tau)\big) - F\big(\alpha_{j-1} - V_i(\beta, \tau)\big).$$

If F were known, the population log-likelihood would be:

$$\ell(\alpha, \beta, \tau; F) = \mathbb{E}\left[\sum_{j=0}^{J} 1_{\{Y_i = j\}} \log p_j(X_i, D_i; \alpha, \beta, \tau, F)\right].$$

In a sample of size n, the infeasible sample log-likelihood is:

$$\ell_n(\alpha, \beta, \tau; F) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=0}^{J} 1_{\{Y_i = j\}} \log p_j(X_i, D_i; \alpha, \beta, \tau, F).$$

When F is unknown, we replace it with a nonparametric estimator $\hat{F}$ constructed via KDE, obtaining the *quasi*-log-likelihood:

$$\hat{\ell}_n(\alpha, \beta, \tau) = \frac{1}{n} \sum_{i=1}^{n} \hat{m}(X_i) \sum_{j=0}^{J} 1_{\{Y_i = j\}} \log \hat{p}_j(X_i, D_i; \alpha, \beta, \tau),$$

where

$$\hat{p}_j(X_i, D_i; \alpha, \beta, \tau) = \hat{F}\big(\alpha_j - V_i(\beta, \tau)\big) - \hat{F}\big(\alpha_{j-1} - V_i(\beta, \tau)\big),$$

and $\hat{m}(X_i)$ is a trimming function that downweights observations in extreme regions of the covariate space where the KDE may be unstable. The KDE-based estimator $(\hat{\alpha}, \hat{\beta}, \hat{\tau})$ is defined as any maximizer of $\hat{\ell}_n(\alpha, \beta, \tau)$.

C.1.2. Kernel Estimation of the Error Distribution

To construct $\hat{F}$, we exploit the relationship between the distribution of the index V_i and the error term ε_i. Let:

$$g_1(v \mid Y \le j) = \text{density of } V \text{ given } Y \le j, \qquad g_0(v \mid Y > j) = \text{density of } V \text{ given } Y > j,$$

and let:

$$\pi_j = \mathbb{P}(Y_i \le j), \qquad 1 - \pi_j = \mathbb{P}(Y_i > j).$$

Then, by Bayes' rule:

$$\mathbb{P}(Y_i \le j \mid V_i = v) = \frac{\pi_j g_1(v \mid Y \le j)}{\pi_j g_1(v \mid Y \le j) + (1 - \pi_j) g_0(v \mid Y > j)}.$$

Since $Y_i^* = V_i + \varepsilon_i$, and $Y_i \le j$ iff $Y_i^* \le \alpha_j$, we have:

$$\mathbb{P}(Y_i \le j \mid V_i = v) = \mathbb{P}\big(\varepsilon_i \le \alpha_j - v\big) = F\big(\alpha_j - v\big).$$

Thus, if we can estimate π_j, $g_1(\cdot)$, and $g_0(\cdot)$ from the data, we can construct an estimator $\hat{F}$ such that:

$$\hat{F}(\alpha_j - v) \approx \widehat{\mathbb{P}}(Y_i \le j \mid V_i = v).$$

The probabilities π_j and $1 - \pi_j$ can be estimated by sample proportions:

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1_{\{Y_i \le j\}}, \qquad 1 - \hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1_{\{Y_i > j\}}.$$

The conditional densities g_1 and g_0 are estimated using KDE. For concreteness, suppose we use a univariate kernel $K(\cdot)$ (e.g., Gaussian) and bandwidths that may depend on j. Let $n_1(j) = \sum_{i=1}^{n} 1_{\{Y_i \le j\}}$ and $n_0(j) = \sum_{i=1}^{n} 1_{\{Y_i > j\}}$. Define the subsamples:

$$\{V_i : Y_i \le j\}, \qquad \{V_i : Y_i > j\},$$

and estimate the conditional densities as:

$$\hat{g}_1(v \mid Y \le j) = \frac{1}{n_1(j) h_{1j}} \sum_{i:\, Y_i \le j} K\left(\frac{v - V_i}{h_{1j}}\right),$$

$$\hat{g}_0(v \mid Y > j) = \frac{1}{n_0(j) h_{0j}} \sum_{i:\, Y_i > j} K\left(\frac{v - V_i}{h_{0j}}\right),$$

where h_{1j} and h_{0j} are bandwidths chosen according to standard rules. Then, the estimated conditional probability is:

$$\widehat{\mathbb{P}}(Y_i \le j \mid V_i = v) = \frac{\hat{\pi}_j \hat{g}_1(v \mid Y \le j)}{\hat{\pi}_j \hat{g}_1(v \mid Y \le j) + (1 - \hat{\pi}_j) \hat{g}_0(v \mid Y > j)}.$$

We interpret this as an estimator of the CDF of ε_i evaluated at $\alpha_j - v$:

$$\hat{F}(\alpha_j - v) \equiv \widehat{\mathbb{P}}(Y_i \le j \mid V_i = v).$$

In practice, the KDE is implemented with local smoothing and trimming to reduce boundary bias and instability in regions with sparse data. Following **?** and **?**, one can use pilot density estimates to define location-specific bandwidths and damping functions. We adopt these techniques, as in **?**, to ensure $\sqrt{n}$-consistency.

C.1.3. Consistency of the KDE-Based Estimator

This section sketches the main consistency result for the KDE-based estimator.

Assumption C.1 (Smoothness of the Index Distribution). *The conditional density of the index $V_i(\beta, \tau)$ given (X_i, D_i) exists, is strictly positive on its support, and is continuously differentiable up to a sufficiently high order. The support of V_i is compact or can be truncated to a compact set via trimming without affecting the asymptotic behavior of the estimator.*

Assumption C.2 (KDE Regularity Conditions). *The kernel $K(\cdot)$ is a bounded, symmetric density with compact support. Bandwidths h_{1j} and h_{0j} satisfy $h_{1j} \to 0$, $h_{0j} \to 0$, and $nh_{1j} \to \infty$, $nh_{0j} \to \infty$ as $n \to \infty$, with additional conditions ensuring the consistency of locally smoothed KDEs.*

Under these conditions, the KDE estimators $\hat{g}_1$ and $\hat{g}_0$ converge uniformly to g_1 and g_0, and the estimated CDF $\hat{F}$ converges uniformly to the true F on compact subsets of the support.

Theorem C.1 (Consistency of the KDE-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions C.1–C.2 hold. Then, as $n \to \infty$,*

$$\left|(\hat{\alpha}, \hat{\beta}, \hat{\tau}) - (\alpha, \beta, \tau)\right| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$.

Because the ordinal model is identified only up to scale, we impose the error-variance normalization $\mathrm{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance $\hat{\sigma}_\varepsilon^2$.

C.2. Normalizing Flows

This appendix provides additional details for the normalizing-flow-based estimator described in Section **??**.

C.2.1. Change-of-Variables Formula

Let Z be a random variable with a known base density $f_Z(z)$ (the standard normal density), and let $T_\theta : \mathbb{R} \to \mathbb{R}$ be an invertible, differentiable transformation parameterized by θ. Define:

$$\varepsilon = T_\theta(Z).$$

Then, by the change-of-variables formula, the density of ε is:

$$f_\theta(\varepsilon) = f_Z\big(T_\theta^{-1}(\varepsilon)\big) \left| \frac{d}{d\varepsilon} T_\theta^{-1}(\varepsilon) \right|.$$

The corresponding CDF F_θ is:

$$F_\theta(u) = \int_{-\infty}^{u} f_\theta(t)\, dt.$$

C.2.2. Ordinal Likelihood with a Flow-Based Error Distribution

We embed the flow-based error distribution into the latent outcome model:

$$Y_i^* = f(X_i, \beta) + D_i^\top \tau + \varepsilon_i, \qquad \varepsilon_i = T_\theta(Z_i), \quad Z_i \sim \mathcal{N}(0,1),$$

with thresholds α defining the observed ordinal outcome:

$$Y_i = j \quad \Longleftrightarrow \quad \alpha_{j-1} < Y_i^* \leq \alpha_j.$$

Let f_θ and F_θ denote the density and CDF of ε_i implied by the flow T_θ. Conditional on (X_i, D_i), the probability of category j is:

$$\mathbb{P}(Y_i = j \mid X_i, D_i; \alpha, \beta, \tau, \theta) = F_\theta\big(\alpha_j - V_i(\beta, \tau)\big) - F_\theta\big(\alpha_{j-1} - V_i(\beta, \tau)\big),$$

where $V_i(\beta, \tau) = f(X_i, \beta) + D_i^\top \tau$. The sample log-likelihood is:

$$\ell_n(\alpha, \beta, \tau, \theta) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=0}^{J} 1_{\{Y_i = j\}} \log \left[F_\theta\big(\alpha_j - V_i(\beta, \tau)\big) - F_\theta\big(\alpha_{j-1} - V_i(\beta, \tau)\right].$$

We estimate $(\alpha, \beta, \tau, \theta)$ jointly by maximizing $\ell_n(\alpha, \beta, \tau, \theta)$ with respect to all parameters.

C.2.3. Consistency of the Flow-Based Estimator

Let $(\alpha_0, \beta_0, \tau_0, F_0)$ denote the true parameters and error distribution.

Assumption C.3 (Flow Approximation and Identification). *The flow family $\{F_\theta : \theta \in \Theta\}$ is rich enough that there exists $\theta_0 \in \Theta$ such that $F_{\theta_0} = F_0$ (or F_{θ_0} approximates F_0 arbitrarily well). The parameter vector $(\alpha, \beta, \tau, \theta)$ is identified up to a positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$.*

Assumption C.4 (Regularity Conditions). *The parameter space for $(\alpha, \beta, \tau, \theta)$ is compact or can be restricted to a compact subset; the log-likelihood function $\ell_n(\alpha, \beta, \tau, \theta)$ satisfies standard regularity conditions (continuity, differentiability, integrability); and the model is correctly specified.*

Theorem C.2 (Consistency of the Flow-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions C.3–C.4 hold. Then, as $n \to \infty$,*

$$\big|(\hat{\alpha}, \hat{\beta}, \hat{\tau}, \hat{\theta}) - (\alpha_0, \beta_0, \tau_0, \theta_0)\big| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha_0, c\beta_0, c\tau_0)$ with $c > 0$.

We impose the error-variance normalization $\mathrm{Var}(\varepsilon_i) = 1$ ex post. Given $\hat{\theta}$, we estimate:

$$\hat{\sigma}_\varepsilon^2 = \mathrm{Var}_{F_{\hat{\theta}}}(\varepsilon_i) \approx \frac{1}{M} \sum_{m=1}^{M} \big(T_{\hat{\theta}}(Z_m)\big)^2,$$

where $Z_m \sim \mathcal{N}(0,1)$ are independent draws. We then report:

$$\hat{\tau}^{\text{unit-var}} = \frac{\hat{\tau}}{\hat{\sigma}_\varepsilon}, \qquad \hat{\beta}^{\text{unit-var}} = \frac{\hat{\beta}}{\hat{\sigma}_\varepsilon}.$$

D. Comprehensive Simulation Details Full Results

D.1. Detailed Specifications of the Seven Data-Generating Processes

We evaluate finite-sample performance across seven distinct DGPs. In each setting, treatment $D_i \sim \text{Bernoulli}(0.5)$ is randomized, and baseline covariates $X_i = (X_{i1}, X_{i2})^\top$ are drawn from independent standard normal distributions. The latent thresholds partition the outcome space into $J + 1 = 5$ ordered categories: $\alpha = (-1.5, -0.5, 0.5, 1.5)^\top$.

a. **Normal Linear (Benchmark):** The classic ordered probit setting.

$$\begin{aligned} Y_i^* &= 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i, \\ \varepsilon_i &\sim \mathcal{N}(0, 1). \end{aligned}$$

b. **Logistic Linear (Benchmark):** The classic ordered logit setting.

$$\begin{aligned} Y_i^* &= 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i, \\ \varepsilon_i &\sim \text{Logistic}(0, 1). \end{aligned}$$

c. **Skewed Lognormal:** Introduces heavily right-skewed latent error.

$$\begin{aligned} Y_i^* &= 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i, \\ \varepsilon_i &= \exp(u_i) - \mathbb{E}[\exp(u_i)], \quad u_i \sim \mathcal{N}(0, 0.5^2). \end{aligned}$$

d. **Polarized Mixture (Bimodal):** Models polarized latent attitudes via a bimodal mixture of normals.

$$\begin{aligned} Y_i^* &= 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \varepsilon_i, \\ \varepsilon_i &\sim 0.5\mathcal{N}(-1.2, 0.6^2) + 0.5\mathcal{N}(1.2, 0.6^2). \end{aligned}$$

e. **Heteroskedastic:** Violates the proportional odds / constant scale assumption.

$$\begin{aligned} Y_i^* &= 0.5X_{i1} - 0.25X_{i2} + 0.5D_i + \sigma_i\varepsilon_i, \\ \sigma_i &= \exp(0.3D_i + 0.2X_{i1}), \quad \varepsilon_i \sim \mathcal{N}(0, 1). \end{aligned}$$

f. **Nonlinear Moderates:** Evaluates nonlinear latent response functions.

$$\begin{aligned} Y_i^* &= 0.5\sin(\pi X_{i1}) - 0.25X_{i2}^2 + 0.5D_i + \varepsilon_i, \\ \varepsilon_i &\sim \mathcal{N}(0, 1). \end{aligned}$$

g. **High Dimensional:** Evaluates performance with 10 covariates, of which only the first four are active, under a standard normal error distribution.

D.2. Full Simulation Results Table

Table **??** provides the complete simulation results evaluating the Wasserstein distance from the estimated conditional distribution to the true distribution across all sample sizes ($N \in \{500, 1000, 2000\}$).

TABLE D.1. Full Simulation Results: Wasserstein Distance to Truth Across All Settings

DGP Setting	N	Estimator	True	Mean	SD	Bias	RMSE
Heteroskedastic	500	Empirical	0.6049	0.6220	0.0811	0.01710	0.0825
		Ordered Probit	0.6049	0.1908	0.1138	-0.41409	0.4293
		Ordered Logit	0.6049	0.2309	0.1227	-0.37397	0.3934
		Structured Flow	0.6049	0.5826	0.1057	-0.02233	0.1075
		Model-Free Flow	0.6049	0.6077	0.0914	0.00278	0.0910
	1000	Empirical	0.6049	0.6173	0.0629	0.01247	0.0638
		Ordered Probit	0.6049	0.1654	0.0801	-0.43945	0.4466
		Ordered Logit	0.6049	0.2033	0.0874	-0.40157	0.4109
		Structured Flow	0.6049	0.5902	0.0821	-0.01465	0.0830
		Model-Free Flow	0.6049	0.6058	0.0582	0.00091	0.0579
	2000	Empirical	0.6049	0.6096	0.0397	0.00472	0.0398
		Ordered Probit	0.6049	0.1801	0.0592	-0.42478	0.4288
		Ordered Logit	0.6049	0.2208	0.0640	-0.38403	0.3893
		Structured Flow	0.6049	0.6038	0.0355	-0.00104	0.0354
		Model-Free Flow	0.6049	0.6045	0.0395	-0.00036	0.0393
High Dimensional	500	Empirical	0.7011	0.7160	0.1055	0.01483	0.1061
		Ordered Probit	0.7011	0.7052	0.1052	0.00409	0.1047
		Ordered Logit	0.7011	0.6903	0.1052	-0.01085	0.1052
		Structured Flow	0.7011	0.6640	0.1416	-0.03718	0.1458
		Model-Free Flow	0.7011	0.7044	0.1001	0.00324	0.0996
	1000	Empirical	0.7011	0.7008	0.0749	-0.00030	0.0745
		Ordered Probit	0.7011	0.6970	0.0692	-0.00412	0.0690
		Ordered Logit	0.7011	0.6822	0.0701	-0.01892	0.0723
		Structured Flow	0.7011	0.6698	0.0774	-0.03130	0.0831
		Model-Free Flow	0.7011	0.7016	0.0724	0.00046	0.0721
	2000	Empirical	0.7011	0.7078	0.0497	0.00669	0.0499
		Ordered Probit	0.7011	0.6989	0.0409	-0.00220	0.0408
		Ordered Logit	0.7011	0.6830	0.0407	-0.01815	0.0443
		Structured Flow	0.7011	0.6980	0.0516	-0.00314	0.0514
		Model-Free Flow	0.7011	0.6983	0.0458	-0.00288	0.0456
Logistic Linear	500	Empirical	0.4930	0.4712	0.1021	-0.02182	0.1039
		Ordered Probit	0.4930	0.4918	0.0836	-0.00125	0.0832
		Ordered Logit	0.4930	0.4924	0.0832	-0.00066	0.0827
		Structured Flow	0.4930	0.4965	0.0921	0.00345	0.0918
		Model-Free Flow	0.4930	0.5028	0.0978	0.00972	0.0978
	1000	Empirical	0.4930	0.4901	0.0709	-0.00294	0.0706
		Ordered Probit	0.4930	0.5102	0.0556	0.01718	0.0579
		Ordered Logit	0.4930	0.5097	0.0554	0.01669	0.0576
		Structured Flow	0.4930	0.5111	0.0685	0.01808	0.0705
		Model-Free Flow	0.4930	0.5110	0.0618	0.01797	0.0640
	2000	Empirical	0.4930	0.4923	0.0549	-0.00075	0.0546
		Ordered Probit	0.4930	0.5044	0.0427	0.01138	0.0440
		Ordered Logit	0.4930	0.5055	0.0415	0.01251	0.0431
		Structured Flow	0.4930	0.5039	0.0476	0.01089	0.0486
		Model-Free Flow	0.4930	0.5047	0.0521	0.01170	0.0531
Nonlinear Moderates	500	Empirical	0.7348	0.7374	0.1066	0.00254	0.1061
		Ordered Probit	0.7348	0.7234	0.1067	-0.01141	0.1068
		Ordered Logit	0.7348	0.7364	0.1056	0.00159	0.1051
		Structured Flow	0.7348	0.7303	0.1255	-0.00454	0.1250
		Model-Free Flow	0.7348	0.7386	0.0973	0.00373	0.0969
	1000	Empirical	0.7348	0.7516	0.0795	0.01678	0.0809
		Ordered Probit	0.7348	0.7337	0.0786	-0.00111	0.0782
		Ordered Logit	0.7348	0.7421	0.0783	0.00730	0.0783
		Structured Flow	0.7348	0.7534	0.0952	0.01862	0.0966
		Model-Free Flow	0.7348	0.7437	0.0700	0.00885	0.0702

E. Empirical Application Details Full Tables

This section reports the full set of empirical results corresponding to the two applications discussed in Section ??: ? and ?. Standard errors are calculated using $B = 500$ bootstrap replications and are presented in parentheses.

E.1. ?: Persuasive Effects of Authoritarian Propaganda

Table ?? reports the point estimates and bootstrapped standard errors for average marginal category-specific effects and cumulative effects across the five candidate estimators.

TABLE E.1. Empirical Application Estimates: Support for Political Model (?)

Metric / Category	Empirical	Ordered Probit	Ordered Logit	Structured Flow	Model-Free Flow
Wasserstein Unit	1.073 (0.038)	0.974 (0.048)	0.986 (0.047)	0.917 (0.116)	1.011 (0.246)
Category Effects (AME)					
Category 1	-0.138 (0.011)	-0.171 (0.007)	-0.166 (0.006)	-0.113 (0.024)	-0.092 (0.050)
Category 2	-0.135 (0.010)	-0.095 (0.006)	-0.113 (0.006)	-0.120 (0.012)	-0.137 (0.028)
Category 3	-0.085 (0.013)	0.015 (0.001)	0.013 (0.002)	-0.055 (0.016)	-0.078 (0.015)
Category 4	0.152 (0.011)	0.071 (0.003)	0.086 (0.004)	0.098 (0.014)	0.063 (0.025)
Category 5	0.110 (0.009)	0.075 (0.004)	0.086 (0.004)	0.096 (0.014)	0.105 (0.022)
Category 6	0.097 (0.008)	0.105 (0.006)	0.094 (0.005)	0.094 (0.031)	0.139 (0.029)
Cumulative Effects					
$P(Y \geq 2)$	0.138 (0.011)	0.171 (0.007)	0.166 (0.006)	0.113 (0.024)	0.092 (0.050)
$P(Y \geq 3)$	0.273 (0.013)	0.266 (0.012)	0.279 (0.012)	0.233 (0.026)	0.229 (0.074)
$P(Y \geq 4)$	0.359 (0.012)	0.251 (0.013)	0.267 (0.014)	0.288 (0.023)	0.307 (0.070)
$P(Y \geq 5)$	0.206 (0.010)	0.180 (0.010)	0.180 (0.010)	0.190 (0.024)	0.244 (0.049)
$P(Y \geq 6)$	0.097 (0.008)	0.105 (0.006)	0.094 (0.005)	0.094 (0.031)	0.139 (0.029)

E.2. ?: Human Rights and Public Support for Intervention

Table ?? reports the results for the replication of ?, indicating consistency across the parametric and flexible flow-based estimators.

TABLE E.2. Empirical Application Estimates: Support for Strike (?)

Metric / Category	**Empirical**	**Ordered Probit**	**Ordered Logit**	**Structured Flow**	**Model-Free Flow**
Wasserstein Unit	0.513 (0.065)	0.500 (0.058)	0.507 (0.059)	0.504 (0.107)	0.509 (0.077)
Category Effects (AME)					
Category 1	0.084 (0.017)	0.081 (0.009)	0.078 (0.009)	0.073 (0.018)	0.068 (0.018)
Category 2	0.068 (0.022)	0.065 (0.008)	0.071 (0.009)	0.080 (0.019)	0.092 (0.021)
Category 3	0.017 (0.020)	0.015 (0.002)	0.016 (0.002)	0.019 (0.017)	0.023 (0.019)
Category 4	-0.061 (0.025)	-0.048 (0.006)	-0.053 (0.007)	-0.065 (0.022)	-0.084 (0.025)
Category 5	-0.108 (0.022)	-0.113 (0.013)	-0.113 (0.013)	-0.106 (0.031)	-0.098 (0.024)
Cumulative Effects					
$P(Y \geq 2)$	-0.084 (0.017)	-0.081 (0.009)	-0.078 (0.009)	-0.073 (0.018)	-0.068 (0.018)
$P(Y \geq 3)$	-0.152 (0.024)	-0.146 (0.017)	-0.150 (0.017)	-0.153 (0.032)	-0.160 (0.028)
$P(Y \geq 4)$	-0.169 (0.025)	-0.161 (0.019)	-0.166 (0.019)	-0.172 (0.038)	-0.183 (0.027)
$P(Y \geq 5)$	-0.108 (0.022)	-0.113 (0.013)	-0.113 (0.013)	-0.106 (0.031)	-0.098 (0.024)